1. Frontend — making it user-friendly

- Group fields logically (location/radius together, budget/duration together) — already done in the layout
- Use placeholder examples in every field so users know the expected format
- Show the place-selection list with checkboxes + short descriptions, not just names — people choose faster when they can see what a place is
- Disable "Generate Itinerary" until required fields are filled, so users don't submit incomplete data
- Show a loading state while itinerary is generating (even a simple spinner) so it doesn't feel stuck
- Keep results scannable — day-wise cards, clear headers, no walls of text

2. Logic — selecting & sequencing places

- We don't use AI/ML — it's a **rule-based heuristic algorithm**:
  - Sort user-selected places by distance from current location (nearest-first)
  - For each day, keep adding the nearest place as long as it fits within the day's time budget (we assume ~8 hrs/day of touring) and the trip's overall budget
  - If a place doesn't fit (budget/time/season), it goes to the "excluded" list with a reason attached
  - This is a **greedy algorithm**, not a perfect optimizer — good enough for our scope, easy to explain in viva

3. Data — sourcing, volume, types

- **Source:** manually curated by us (researched from real info — Google, tourism sites, etc.), possibly supplemented by a public dataset (Kaggle) or OpenStreetMap for basic place listings
- **Volume:** scoped to **one state/region**, ~20-25 places — enough to be a real, usable demo without an unmanageable research load
- **Types of data needed per place:** name, coordinates (lat/lng), category, short description, average visit duration, entry cost, whether it needs prior permission (+ details), best season to visit
- Separately: a small curated list of hotels and food options per place/region, with rough cost ranges

4. Maps — Google Maps or alternative

- Recommending **Leaflet.js** + **OpenStreetMap** instead of Google Maps
- Reason: Google Maps JS API needs a Google Cloud **billing account** set up (card on file) even though our usage will stay in the free tier — unnecessary setup friction for a student project
- Leaflet + OpenStreetMap needs **no API key, no billing, no approval** — just works, and is free to use without limits for our scale

5. Logic — how "pace" affects the output

- Pace controls how many hours per day we assume for touring, which changes how many places fit per day:
  - **Relaxed:** ~5-6 hrs/day → fewer places/day, more rest time built in
  - **Moderate:** ~7-8 hrs/day → balanced default
  - **Packed:** ~9-10 hrs/day → more places squeezed per day, less buffer time
- This directly feeds into the Day 2 allocation logic — it's just a variable (`MAX_DAY_HOURS`) that changes based on the selected pace, nothing more complex than that